Introduction to Programming

Lecture 15

VU
Virtual University

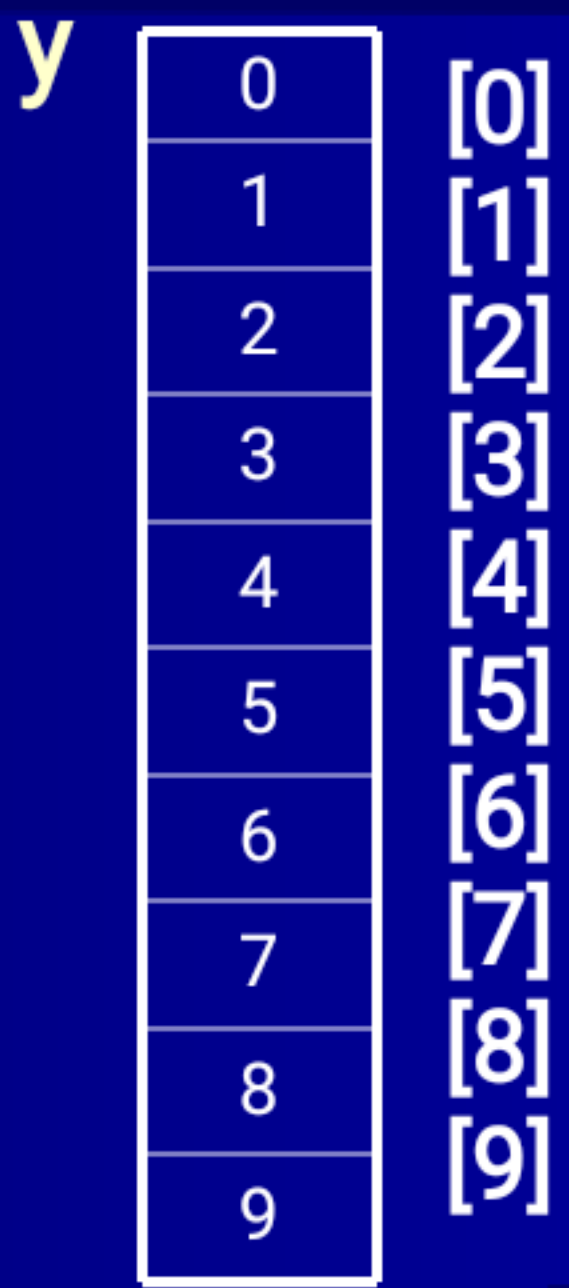
In Today's Lecture

- Pointers and Arrays Manipulations
- Pointers Expression
- Pointers Arithmetic
- Multidimensional Arrays
- Pointer String and Array

Pointers and Arrays

```
int y[10];
```

Starting Address of Array



The name of the array is like a **pointer** which contain the address of the first element.

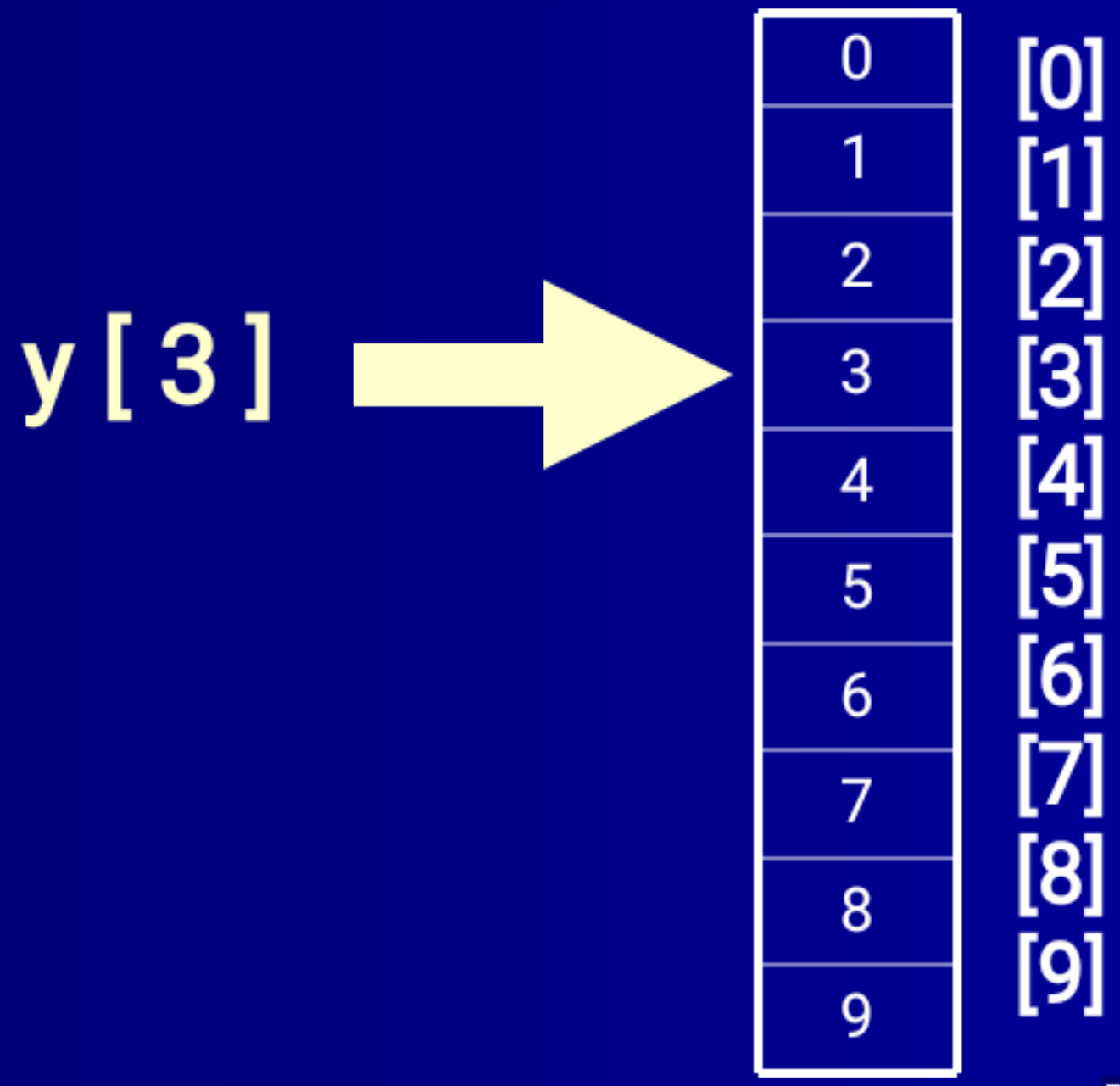
Declaration of a Pointer Variable

```
int y [ 10 ] ;
int *yptr ;
```

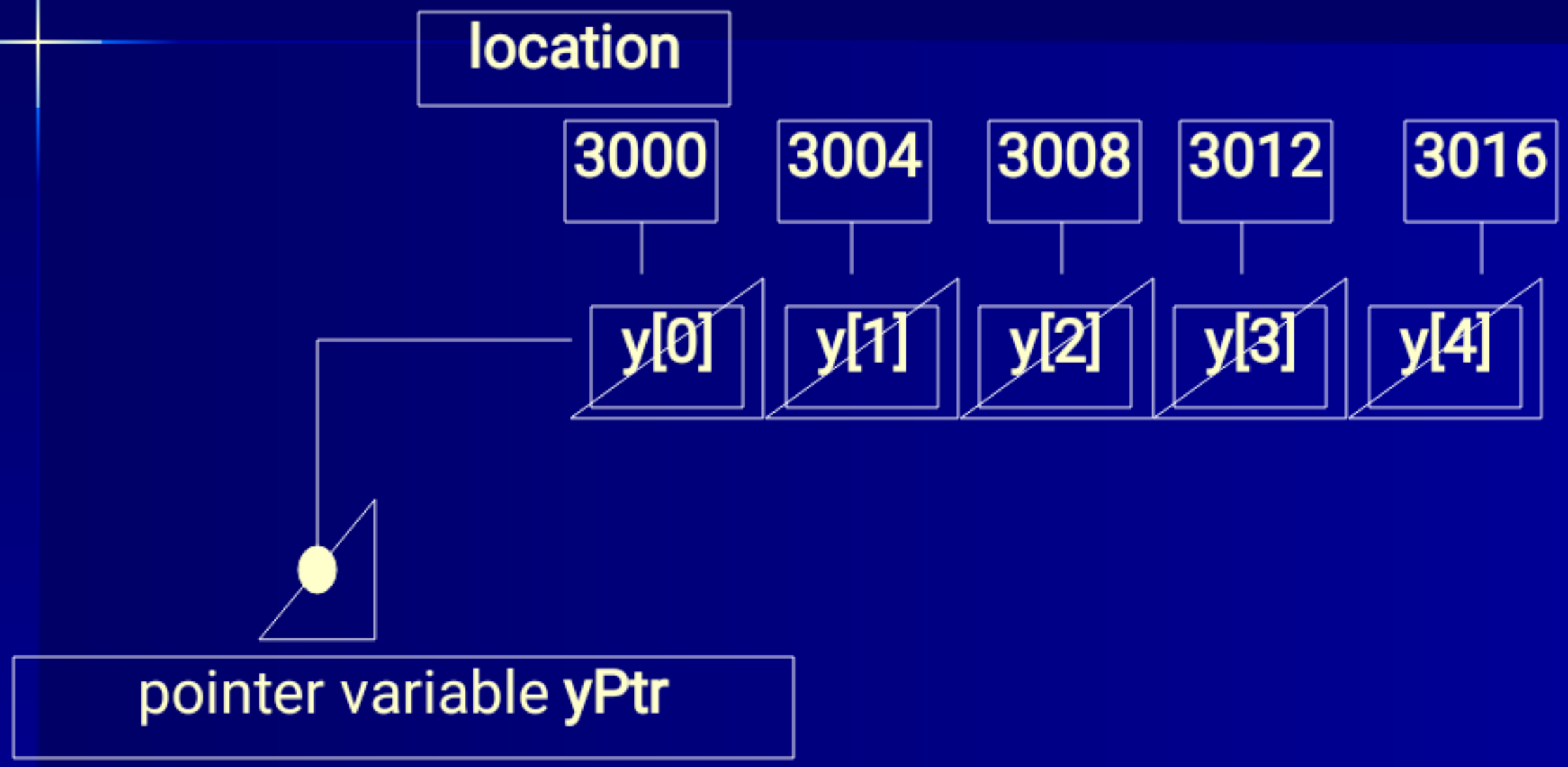
yptr is a pointer to integer

```
yptr = y ;
```

Declaration of a Pointer Variable



```
int y [ 10 ] ;
int *yptr ;
yptr = y ;
yptr ++ ;
```


location
3000
3004
3008
3012
3016
y[0]
y[1]
y[2]
y[3]
y[4]
pointer variable yPtr

In this case yptr is a pointer to integer so now when we increment yptr it points to the next integer

Example 1

```
#include<iostream.h>
main ( )
{
 int y [ 10 ] ;
 int *yptr = y ;
 yptr = y ;
 cout << yptr ;
 yptr ++ ;
 cout << yptr ;
}
```

```
yptr = y ;
```

is same as

```
yptr = &y [ 0 ] ;
........
yptr = &y [ 2 ] ;
```

Example 2

```
#include<iostream.h>
main ( )
{
 int y [ 10 ] ;
 int *yptr ;
 yptr = y ;
 cout << yptr ;
 yptr ++ ;
 cout << *yptr ;
}
```

Example 3

```
main ( )
{
  int x = 10 ;
  int *yptr ;
  yptr = &x ;
  cout << yptr ;
  cout << *yptr ;
  *yptr ++ ;
}
```

increment whatever yptr points to

Pointer Arithmetic

```
*yptr + 3 ;
```

This Is an Expression

```
cout << *yptr ;
*yptr += 3 ;
```

Pointer Arithmetic

```
yptr = &x ;
yptr ++ ;
```

Pointer Arithmetic

```
int x =10 ;
int *yptr ;
yptr = &x ;
*yptr += 3 ;
yptr += 3 ;
```

Decrementing

*yptr --

Pointer Arithmetic

```
int *p1 ,*p2;
.....
p1 + p2 ;
```

Error

Pointer Arithmetic

```
int y [ 10 ] , *y1 , *y2 ;
        y1 = &y [ 0 ] ;
        y2 = &y [ 3 ] ;
        cout << y2 - y1 ;
```

Pointer Arithmetic

```
int y [ 10 ] ;
int *yptr ;
yptr = y [ 5 ] ;
cout << *( yptr + 5 ) ;
```

Pointer Comparison

```
if ( y1 > y2 )
if ( y1 >= y2 )
if ( y1 == y2 )
```

Pointer Comparison

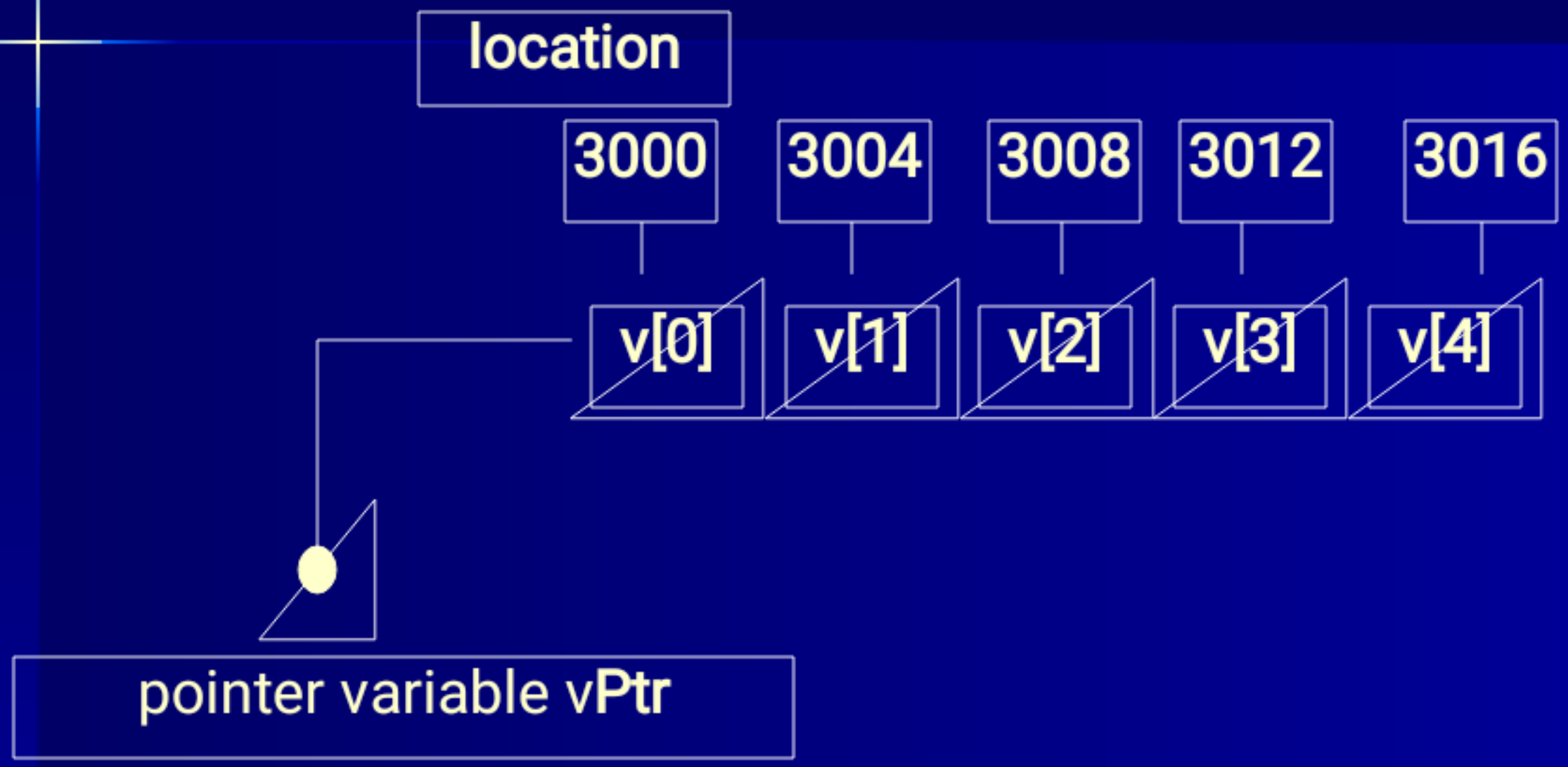
```
if ( *y1 > *y2 )
```

Example

```
int y [ 10 ] ;
int *yptr ;
yptr = y ;
cout << y [ 5 ] ;
cout << ( yptr + 5 ) ;
cout  << *( yptr + 5 ) ;
```

Example

```
int que [ 10 ] ;
int y [ 10 ];
int *yptr ;
yptr = y ;
yptr = que ;
```


location
3000
3004
3008
3012
3016
v[0]
v[1]
v[2]
v[3]
v[4]
pointer variable vPtr

Strings

String Initialization

```
char name [ 20 ] ;
name [ 0 ] = 'A' ;
name [ 1 ] = 'm' ;
name [ 2 ] = 'i' ;
name [ 3 ] = 'r' ;
name [ 4 ] = '\0' ;
```

String Initialization

Strings is always terminated with \0

String Initialization

char name [20] = “Amir” ;

Array must be one character space larger than the number of printable character which are to be stored.

Example 4

```
char string1 [ 20 ] = “Amir”;
char string2 [ 20 ] ;
char *ptrA, *ptrB ;
prtA = string1 ;
prtB = string2 ;
while ( *ptrA != ‘\0’ )
{
 *ptrB++ = *ptrA++;
}
*ptrB = ‘\0’ ;
```

String Copy Function

```
myStringCopy ( char *destination , const char *source )
{
 while ( *source != '\0' )
 {
  *destination++ = *source++ ;
 }
 *destination = '\0' ;
}
```

In Today's Lecture

- Pointers and Arrays
- Pointer Arithmetic
- Manipulation of Arrays
- String Arrays